

SOUVIK ROY

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PROFESSIONAL SUMMARY

Software/Firmware Engineer contributing to Flash R&D and accelerator firmware development at Marvell Technology, with exposure to embedded systems, Linux system programming, and storage technologies including NVMe, PCIe, and NAND Flash. working directly alongside chip-level lab teams. Long-term focus aligned toward AI accelerator architecture and high-performance storage systems.

PROFESSIONAL EXPERIENCE

Marvell Technology | *Software/Firmware Engineer (Contingent via Wipro)* | Pune, Maharashtra **Mar 2025 – Present**

- Contributing to the **Flash R&D** team developing next-generation SSD accelerator firmware for high-performance storage controller.
- Supporting firmware debugging and validation activities across firmware and hardware development workflows.
- Working extensively with **NVMe protocol stacks, PCIe interface layers, and NAND Flash management subsystems** for storage command processing and queue management.
- Collaborating with silicon and chip-level lab teams on firmware integration, bring-up testing, and hardware-software boundary validation.
- Gaining exposure to **accelerator pipeline architecture** and firmware-driven data flow orchestration across SSD storage subsystems.

Wipro Limited | *Project Engineer* | Pune, Maharashtra **Feb 2025 – Present**

- Completed intensive training in **Linux System Programming** — POSIX threads, process management, TCP/IP sockets, synchronization, and file handling in C/C++.
- Applied industry-level software engineering practices across structured debugging exercises and system-level problem-solving scenarios in Linux environments.

TECHNICAL SKILLS

Firmware & Embedded: Embedded Firmware Development, Firmware Debugging, Storage Controller Architecture

Storage Protocols: NVMe, PCIe, NAND Flash Architecture, SSD Subsystems

System Programming: Linux, POSIX Threads, TCP/IP Socket Programming, Multithreading, Mutex Synchronization

Languages & Tools: C, C++ (C++17), Java, Git, GitHub

Additional: HTML, CSS, JavaScript, MySQL

PROJECT WORK

Enterprise Concurrent Signal Broadcast Framework | *Linux · C++17 · TCP/IP · POSIX 2026*

- Architected a **multi-threaded TCP/IP server-client framework** in Linux using POSIX sockets with secure session-based communication and real-time broadcast handling.
- Engineered thread-safe concurrency using **std::thread and mutex synchronization** with low-level socket APIs (socket, bind, listen, accept, recv, send).
- Implemented secure encrypted messaging, session lifecycle management, and centralized logging with process-level resource cleanup.

EDUCATION

University of Engineering & Management, Kolkata — *B.Tech — Computer Science & Engineering 2021 – 2025* | CGPA: 8.3

Pannalal Institution — *Higher Secondary — WBCHSE 2021* | 82%

Bedibhawan Rabitirtha Vidyalaya — *Secondary — WBBSE 2019* | 84%

TECHNICAL INTERESTS

AI Accelerator Architecture · Embedded Firmware Engineering · Flash Storage Systems · NVMe/PCIe Protocol Design · Edge AI & On-Device Inference